

5G Technology

Learning Objectives

- Explain 5G technical characteristics and how they differ from previous generations
- Analyze how 5G enables new technologies and applications
- Evaluate the societal and economic implications of 5G deployment

Engage (5-7 minutes)

Demonstrate latency: ask students to tap a desk as quickly as possible. That's roughly 100 milliseconds. 5G aims for 1 millisecond latency—100 times faster. Show speed comparisons: 3G = walking, 4G = driving, 5G = flying. Ask students: what becomes possible when connectivity is nearly instantaneous?

Explore (10-12 minutes)

Students map 5G capabilities to applications: enhanced mobile broadband (8K streaming, cloud gaming), massive machine-type communications (IoT with millions of devices), and ultra-reliable low-latency communications (remote surgery, autonomous vehicles). They identify which applications are impossible without 5G and discuss deployment challenges (infrastructure cost, coverage gaps, device availability).

Explain (10-12 minutes)

5G is the fifth generation of mobile network technology, offering three transformative capabilities: enhanced mobile broadband (up to 10 Gbps download speeds), massive machine-type communications (supporting 1 million devices per square kilometer), and ultra-reliable low-latency communications (1ms latency for critical applications). 5G uses three spectrum bands: low-band (wide coverage, moderate speed), mid-band (balance of speed and coverage), and millimeter wave (extreme speed, short range). These capabilities enable real-time applications previously impossible with mobile networks.

Elaborate (8-10 minutes)

Examine 5G's role in enabling other emerging technologies: IoT devices require massive connectivity, autonomous vehicles need instant communication, remote surgery demands perfect reliability, and smart cities depend on ubiquitous sensors. Discuss the infrastructure challenge: 5G requires dense networks of small cells, fiber optic backhaul, and massive antenna arrays (MIMO). Address concerns: health debates (scientifically unfounded but persistent), geopolitical tensions

(Huawei controversy), digital divide (urban vs. rural access), and the energy consumption of billions of always-on devices.

Evaluate (5-8 minutes)

Students create an impact assessment for 5G in a specific context (rural healthcare, urban transportation, or disaster response). They analyze: which 5G capability enables each application, infrastructure requirements, cost-benefit analysis, and potential risks. Debate: “5G will transform society more than the internet did”—students argue for and against.